

NEO Mathematics Learner Guide

Ratio and Proportion - Scaling Recipes

Can maths help us share, scale and sustain things fairly?

What are we exploring?

You will explore how quantities stay connected when something becomes bigger or smaller. You will scale a smoothie recipe, build ratio tables and design your own proportional recipe.

Your Stepping Stones

Notice - how quantities are connected.

Find - a scale factor.

Scale - quantities up and down.

Build - a ratio table.

Explain - why a relationship stays proportional.

A possible way through

Start here: Read the Driving Question.

Explore: Try the Recipe Scaler. Change the number of people several times.

Notice: Ask what changes and what stays connected.

Build: Try the Ratio Table Builder.

Pause: Use the Rest section whenever you need it.

Create: Design a smoothie using proportional quantities.

Practise: Open the Practice Companion.

Finish: Create your Portfolio Piece or save where you reached.

You can move backwards and forwards. You do not have to complete every section in one sitting.

When you feel stuck

1. Find the starting amount.
2. Find the new amount.
3. Ask: how many times bigger or smaller is it?
4. Use that multiplier everywhere.
5. Check whether the relationship still looks like the same recipe.

You can also use: **Socratic prompt**, **Hint**, **Check my answer**, **Worked solution** and the **NEO Maths Scratchpad**.

Using support is part of learning. It is not cheating.

What you might need

Useful

- Device
- Paper or notes app
- Calculator if helpful

You might also like

- Measuring jug
- Kitchen scales
- Recipe card or food packaging
- The embedded Ratio Scratchpad. You do not need to make a real smoothie to complete the maths.

How will I know I am making progress?

You may be making progress when you can say:

"I can spot which quantities are connected."

"I can find a multiplier."

"I can scale more than one quantity using the same multiplier."

"I can explain why the ratio stays the same."

You do not need all of these to be true yet. They are signs of where the learning is heading.

Before you finish

One thing I noticed...

One thing that feels clearer...

One thing I may return to...

The support that helped me most was...

Your Scratchpad

The Scratchpad stays inside the lesson. You can draw, annotate, make a prediction, use a grid or clear the board and begin again. You can use paper instead. The Scratchpad is there to help you think.